

Crisis Tweet Classification across Sparse, Recurrent, and Transformer Models

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Abstract

We build a 12-class disaster-type classification pipeline for tweets, using 11,015 English tweets drawn from the CrisisLexT26 crisis-tweet corpus (Olteanu et al., 2015). We compare three classical machine learning models (Logistic Regression, Random Forest, Naive Bayes) against six recurrent neural architectures (SimpleRNN, GRU, LSTM, and their bidirectional variants) and a fine-tuned BERT Base model (Devlin et al., 2019), over TF-IDF and Word2Vec representations, each tuned across three hyperparameter configurations and selected by validation macro-F1. Our best model, fine-tuned BERT Base, reaches 99.83% test macro-F1, with classical Logistic Regression close behind at 99.01%. We treat this extremely high performance as a phenomenon requiring explanation rather than a straightforward success: our exploratory analysis shows that because most of the 12 disaster classes derive from only one or a handful of specific named real-world events, the dominant lexical signal in the corpus is event-identifying vocabulary (place names, event hashtags) rather than general disaster-type language, which substantially inflates all reported scores relative to how these models would perform on tweets from a genuinely new, unseen disaster. We also document a severe and reproducible architectural failure: unidirectional GRU collapses to single-class prediction (macro-F1 = 1.44%) across every tuning configuration, and unidirectional LSTM degrades sharply with increasing model capacity, while their bidirectional counterparts are among the strongest models tested.

1 Introduction

Social media is often the fastest available information channel during a disaster, but a live tweet stream mixes posts from many simultaneous or sequential crisis events. An emer-

gency responder or aid organization monitoring such a stream needs to know, automatically and quickly, *which* disaster a given tweet concerns, so that relevant posts can be routed to the team handling that specific event. This is an established problem in the crisis informatics literature (Imran et al., 2015), and it is the task we address here: given the raw text of a tweet posted during a disaster, predict which of 12 disaster types it refers to (Flood, Earthquake, Wildfire, Typhoon, Transportation Accident, Explosion, Shooting, Bombing, Haze, Meteor, Building Collapse, Fire).

Disaster-type classification from short, informal social-media text is a genuinely non-trivial multi-class problem in principle: several classes share surface-level vocabulary. Explosion, Bombing, and Fire all co-occur with words like *smoke*, *injured*, and *blast*; Flood and Typhoon share evacuation- and rescue-related vocabulary. A classifier relying only on generic disaster language should therefore struggle to cleanly separate these categories.

In practice, every model we trained – from Naive Bayes to fine-tuned BERT – reached macro-F1 scores at or above 92% (with two dramatic exceptions, discussed below), and our best model exceeds 99.8%. Rather than reporting this uncritically as a successful classifier, we treat it as a result that needs explaining, and ground that explanation directly in our exploratory data analysis. This leads to our two guiding research questions:

1. **RQ1:** What property of this corpus allows near-perfect classification on a 12-class task with semantically overlapping

categories, and what does that imply about how these models would perform on tweets from a disaster event not represented in training data?

2. **RQ2:** Among classical, recurrent, and Transformer-based architectures, which are robust to this task’s characteristics, and which fail – and why, in architectural terms?

2 Related Work

Crisis informatics research has established social media, and Twitter specifically, as a valuable real-time channel during emergencies (Imran et al., 2015). The CrisisLexT26 corpus we use (Olteanu et al., 2015) was built to support exactly this line of work: it provides human-annotated tweets from 26 crisis events spanning 2012–2013, labeled for relevance and information type, and has served as a benchmark for crisis-related text classification since its release. We note that the derived dataset used in our own pipeline labels its source only generically; the underlying event structure and tweet content match CrisisLexT26 directly, which we cite here as the authoritative source.

Our text representations follow the standard progression in NLP: sparse TF-IDF vectors and distributed Word2Vec embeddings learned via the skip-gram objective (Mikolov et al., 2013). For sequence modeling we compare the classical recurrent architectures – LSTM (Hochreiter and Schmidhuber, 1997) and GRU (Cho et al., 2014) – against their bidirectional counterparts, and against BERT (Devlin et al., 2019), a Transformer-based (Vaswani et al., 2017) pretrained language model that is the standard strong baseline for text classification generally.

Distinct from prior applications of CrisisLexT26, which typically use it for binary relevance classification or information-type classification, we repurpose it for a 12-way task the corpus was not originally labeled for: distinguishing which specific disaster type a tweet concerns, pooled across all 26 source events. This reframing is, we argue, exactly

Disaster Type	Count	%
Typhoon	1,080	9.80
Transportation Accident	1,072	9.73
Wildfire	1,041	9.45
Earthquake	1,029	9.34
Flood	1,021	9.27
Explosion	951	8.63
Shooting	914	8.30
Bombing	907	8.23
Haze	903	8.20
Meteor	877	7.96
Building Collapse	832	7.55
Fire	388	3.52
Total	11,015	100.0

Table 1: Class distribution (imbalance ratio 2.78×).

what produces the near-perfect scores central to this paper’s discussion – an artifact of how a multi-class task was constructed from a small number of named events, not a property of CrisisLexT26 itself.

3 Methodology

3.1 Dataset

We use **11,015 English tweets** across **12 disaster classes**, derived from CrisisLexT26 (Olteanu et al., 2015). Table 1 shows the class distribution: mild imbalance, with Fire as a clear outlier at 388 rows (it derives from a single source event – the 2013 Brazil nightclub fire – with comparatively few English-language tweets). There are no missing values in any column. Tweets span 14 countries.

On average, a tweet has 113.1 raw characters (std 28.1) and 16.4 raw words (std 5.1) which confirms the corpus is from 2012-13 as the character limit back then was 280. Upon stripping away the retweet markers, URLs and mentions, it was found that 51.9% of the tweets are retweets (RT @...) while 9.0% are exact duplicate text, and 10.5% are near-duplicates, which is quite a substantial fraction. These distributions have been displayed visually in Figure 1 and Figure 2.

Critically for the discussion in Section 4, the corpus’s most frequent unigrams are dominated not by generic disaster vocabulary but by *event- and location-identifying terms*: *bangladesh* (918 occurrences), *philippine* (759), *lax* (729), *boston* (655), *sghaze*

chitectures), each with a Word2Vec-initialized embedding layer, tuned across 3 configurations varying layer width, depth, and learning rate (18 runs).

BERT Base. bert-base-uncased (Devlin et al., 2019), fine-tuned with the HuggingFace Trainer API across 3 configurations varying learning rate, batch size, and epoch count (3 runs), on GPU.

4 Results

4.1 Hyperparameter Tuning

All 30 tuning runs (3 configurations $\times$ 10 model families) are logged in Table 2, reporting exact hyperparameters and validation metrics for every run, per the project requirement to record all tuning runs in a dedicated table.

Two architectural patterns are visible before test-set evaluation. First, every bidirectional recurrent variant substantially outperforms its unidirectional counterpart (SimpleRNN: 0.925 $\rightarrow$ BiSimpleRNN: 0.961; LSTM: 0.166 $\rightarrow$ BiLSTM: 0.939; GRU: 0.014 $\rightarrow$ BiGRU: 0.978). Second, unidirectional GRU fails catastrophically in *every* one of its 3 configurations (macro-F1 between 0.0133 and 0.0144, statistically indistinguishable from one another), and unidirectional LSTM’s best configuration reaches only 0.166 – both far below the random-guess baseline’s neighborhood of informativeness one would expect even a weak classifier to exceed. Third, for classical models, more regularization capacity (higher C, more trees, higher smoothing α) consistently helps within the tested range, suggesting none of the three had yet reached a point of overfitting at the configurations tried.

4.2 Overall Model Comparison

Table 3 reports test-set performance for every model’s best configuration. BERT Base is the strongest model overall; unidirectional GRU is the weakest by a very wide margin.

4.3 Why Are the Scores So High? Grounding in EDA

A 99.8% macro-F1 on a 12-class task with semantically overlapping categories (Section 1)

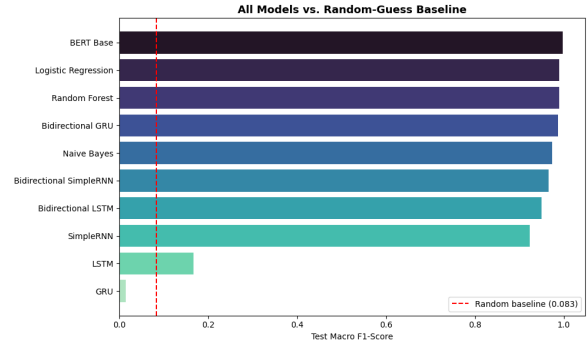


Figure 5: Test macro-F1, all models, against the random-guess baseline ($1/12 \approx 0.083$).

is not, on its face, plausible for a model relying on genuine disaster-type language understanding. Section 3.1’s EDA finding – that the corpus’s most frequent terms are dominated by location/event names (*bangladesh*, *philippine*, *lax*, *boston*, *sghaze*, *colorado*) rather than generic disaster vocabulary – explains why. Because most of our 12 classes derive from only one or a handful of named real-world events (e.g. Haze consists entirely of tweets about the 2013 Singapore haze, identifiable via the single hashtag #sghaze; Bombing is dominated by the Boston Marathon bombing), a classifier can achieve near-perfect accuracy by learning to recognize these event fingerprints rather than transferable disaster-type language. The Word2Vec nearest-neighbor check in Section 3.4 corroborates this at the representation level: *earthquake*’s closest neighbors include *usgs* (a specific agency), and *flood*’s include *calgary* (a specific city) – precisely the kind of location-bound co-occurrence that would not generalize to a disaster in a different country.

This does not mean the reported numbers are meaningless, but it does mean they should not be read as an estimate of how these models would perform on tweets from a genuinely new disaster event that shares no hashtags, city names, or agency handles with the training data – the realistic deployment scenario for a crisis-tweet triage system. We were unable to run a full leakage-stripping ablation (removing hashtags, proper nouns, and location names and retraining, as recommended for future work) within this project’s scope, but the

Model	Configuration	Val Acc.	Val Macro-F1	Train (s)
Logistic Regression	C=0.1, L2	0.9637	0.9627	1.77
Logistic Regression	C=1.0, L2, balanced	0.9843	0.9856	3.76
Logistic Regression	C=5.0, L2	0.9873	0.9882	3.73
Random Forest	n=100, depth=20	0.9619	0.9643	1.01
Random Forest	n=200, depth=None	0.9843	0.9857	5.55
Random Forest	n=300, depth=None, min_split=4	0.9849	0.9862	7.60
Naive Bayes	$\alpha=0.1$	0.9534	0.9560	0.00
Naive Bayes	$\alpha=0.5$	0.9619	0.9633	0.00
Naive Bayes	$\alpha=1.0$	0.9673	0.9686	0.00
SimpleRNN	64 units, LR=1e-3	0.5324	0.4870	11.70
SimpleRNN	128 units, dropout=0.2, LR=5e-4	0.9244	0.9254	13.40
SimpleRNN	2-layer stacked, dropout=0.3, LR=5e-4	0.8518	0.8404	11.40
Bi-SimpleRNN	64 units, LR=1e-3	0.9595	0.9611	24.53
Bi-SimpleRNN	128 units, dropout=0.2, LR=5e-4	0.8953	0.9004	13.53
Bi-SimpleRNN	2-layer stacked, dropout=0.3, LR=5e-4	0.8518	0.8464	20.75
GRU	64 units, LR=1e-3	0.0865	0.0133	7.90
GRU	128 units, dropout=0.2, LR=5e-4	0.0944	0.0144	4.17
GRU	2-layer stacked, dropout=0.3, LR=5e-4	0.0944	0.0144	5.77
Bi-GRU	64 units, LR=1e-3	0.9758	0.9778	13.11
Bi-GRU	128 units, dropout=0.2, LR=5e-4	0.9268	0.9293	6.96
Bi-GRU	2-layer stacked, dropout=0.3, LR=5e-4	0.9014	0.9037	8.90
LSTM	64 units, LR=1e-3	0.1658	0.0471	4.31
LSTM	128 units, dropout=0.2, LR=5e-4	0.2722	0.1548	4.37
LSTM	2-layer stacked, dropout=0.3, LR=5e-4	0.2819	0.1663	6.42
Bi-LSTM	64 units, LR=1e-3	0.9407	0.9392	5.40
Bi-LSTM	128 units, dropout=0.2, LR=5e-4	0.9304	0.9341	6.72
Bi-LSTM	2-layer stacked, dropout=0.3, LR=5e-4	0.9141	0.9172	11.85
BERT Base	LR=2e-5, batch=32, epochs=3	0.9903	0.9907	263.83
BERT Base	LR=3e-5, batch=32, epochs=4, warmup	0.9915	0.9918	353.11
BERT Base	LR=5e-5, batch=32, epochs=3	0.9903	0.9907	265.35

Table 2: Complete hyperparameter tuning log, all 30 runs. Bold marks the configuration selected as optimal for each model family (highest validation macro-F1).

Model	Test Acc.	Test Macro-F1
BERT Base	0.9982	0.9983
Logistic Regression	0.9891	0.9901
Random Forest	0.9897	0.9897
Bidirectional GRU	0.9855	0.9868
Naive Bayes	0.9728	0.9744
Bidirectional SimpleRNN	0.9643	0.9666
Bidirectional LSTM	0.9492	0.9506
SimpleRNN	0.9189	0.9231
LSTM	0.2892	0.1667
GRU	0.0944	0.0144

Table 3: Final test-set comparison, all 10 models (best configuration each), sorted by macro-F1.

Class	P	R	F1
Wildfire (only predicted class)	0.094	1.000	0.173
All 11 other classes	0.000	0.000	0.000
Macro avg	0.008	0.083	0.014

Table 4: GRU (worst model): condensed test-set report. Full 12-row report in Appendix A.

4.4 Architectural Failure Analysis: GRU and LSTM

Table 4 (full report in Appendix A) shows the unidirectional GRU’s failure is total and highly specific: 0.00 F1 on 11 of 12 classes, with 100% recall and 9.4% precision on the remaining class (Wildfire) – the textbook signature of a model that predicted a single class for every test example, having failed to learn anything beyond the training set’s marginal class frequency. Unidirectional LSTM (Appendix A) shows a related but less total failure: it achieves reasonable recall on 3-4 classes (Haze, Transportation Accident, Shooting) at very poor precision, while collapsing to zero on the remaining 8 – a partial version of the same failure mode.

We attribute both failures to the vanishing/exploding-gradient sensitivity

qualitative evidence above – both the raw corpus statistics and the Word2Vec embedding space – strongly suggests that a substantial fraction of the headline performance in Table 3 reflects event memorization rather than generalizable disaster-type classification. We flag this as the single most important caveat for interpreting this paper’s results, and recommend any deployment or extension of this pipeline first quantify the gap directly via such an ablation.

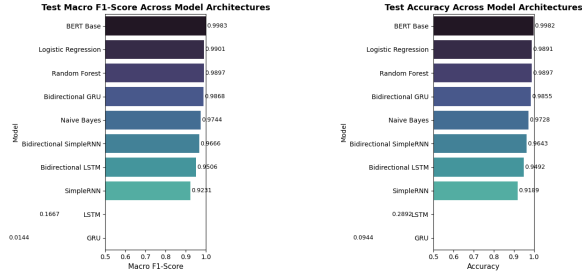


Figure 6: Comparative performance, all models (accuracy and macro-F1).

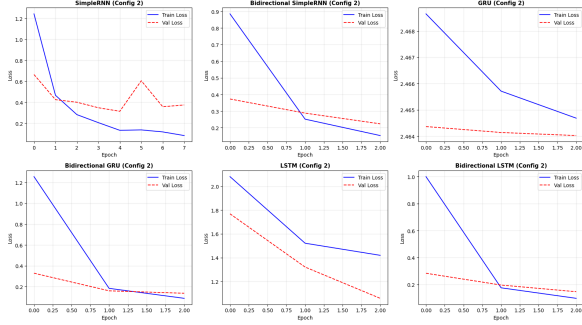


Figure 7: Training/validation loss curves, all 6 recurrent architectures.

well documented for unidirectional recurrent architectures (Hochreiter and Schmidhuber, 1997), which is mostly gone after a bidirectional wrapper is added: every bidirectional variant exceeds 96% test macro-F1, including Bidirectional GRU (98.7%) and Bidirectional LSTM (95.1%), thus, it becomes clear that bidirectional models performed much better in similar tests. Figure 6 and Figures 7–8 show this pattern across all six recurrent architectures.

4.5 Per-Class Performance Across Models

Table 5 breaks down test F1-score by class for the nine models with a full classification report available (Random Forest’s per-class breakdown was not separately logged in the source notebook; only its aggregate accuracy/macro-F1 appear in Table 3). Meteor, Haze, Building Collapse, and Fire – the classes with the narrowest, most distinctive vocabulary per Figure 4 – score at or near 1.00 F1 for every model that has not collapsed entirely, including the otherwise-weak LSTM. Flood is the single hardest class for most mod-

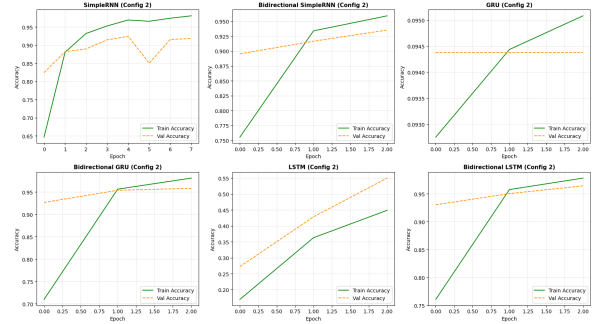


Figure 8: Training/validation accuracy curves, all 6 recurrent architectures.

els: it is Bidirectional LSTM’s worst class ($F1 = 0.855$) and Logistic Regression’s second-worst ($F1 = 0.968$), consistent with the t-SNE overlap noted below and with Flood being one of the few classes (with Earthquake and Transportation Accident) that spans multiple real-world source events rather than a single one, giving it a more varied, less fingerprint-able vocabulary.

4.6 Representation Quality and Confusion Patterns

Figure 9 projects the TF-IDF space to two dimensions via truncated SVD followed by t-SNE. Classes with unique and limited vocabulary (Meteor, Haze, Building Collapse) are tightly clustered together, resulting in near-perfect per-class F1 scores in Table 5; Flood shows the highest overlap with neighboring clusters, resulting in getting the lowest-scoring class for several models noted above. Figure 10 and Figure 11 show confusion matrices for all 10 models.

4.7 Bonus: Ensemble and Real-Time Inference

A weighted soft-voting ensemble combining the strongest models reaches 99.82% test accuracy and 99.83% macro-F1 – matching but not exceeding BERT Base alone, consistent with the ensemble’s members already agreeing on the overwhelming majority of test examples. We also implemented a real-time triage inference function that takes a raw, unseen tweet and returns a predicted disaster type with confidence score; qualitative spot checks on hand-written example tweets (not

Class	BERT	LogReg	NaiveB	RNN	GRU	LSTM	BiRNN	BiGRU	BiLSTM
Bombing	0.996	0.978	0.968	0.863	0.000	0.069	0.931	0.975	0.947
Building Collapse	1.000	1.000	0.992	0.965	0.000	0.000	1.000	1.000	0.980
Earthquake	1.000	0.994	0.984	0.947	0.000	0.000	0.977	0.990	0.945
Explosion	0.997	0.979	0.971	0.910	0.000	0.000	0.954	0.983	0.946
Fire	1.000	1.000	0.983	0.966	0.000	0.000	0.991	1.000	0.959
Flood	0.994	0.968	0.918	0.835	0.000	0.025	0.940	0.968	0.855
Haze	1.000	1.000	1.000	0.959	0.000	0.895	0.963	0.993	0.982
Meteor	1.000	1.000	1.000	0.977	0.000	0.000	1.000	1.000	0.996
Shooting	1.000	1.000	0.993	0.938	0.000	0.343	0.982	0.996	0.971
Transportation Accident	0.997	0.991	0.975	0.922	0.000	0.479	0.960	0.984	0.954
Typhoon	0.997	0.975	0.951	0.898	0.000	0.000	0.946	0.972	0.936
Wildfire	1.000	0.997	0.959	0.898	0.173	0.189	0.954	0.981	0.935

Table 5: Per-class test F1-score, all models with a logged full classification report (Random Forest excluded; see text). RNN = SimpleRNN, BiRNN = Bidirectional SimpleRNN.

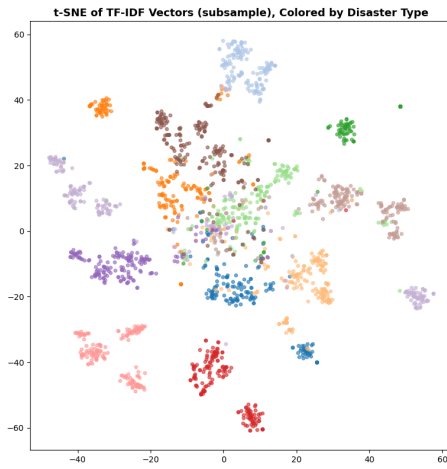


Figure 9: t-SNE of TF-IDF vectors (subsample), colored by disaster type.

drawn from the corpus) produced plausible top predictions (e.g. an earthquake tweet correctly triaged with 79.8% confidence, a flood tweet with 86.1% confidence), though we note these hand-written examples still use natural disaster-descriptive vocabulary and do not test the harder case of a tweet from an entirely new named event, which Section 4.3 suggests would be the more demanding real-world test.

5 Conclusion

We built a complete disaster-type classification pipeline based on a dataset containing English crisis tweets. It has 11,015 rows spanning 12 classes. We tested 3 classical models, 6 recurrent architectures, and a fine-tuned BERT Base model, each tuned across 3 hyperparameter configurations. Our best model, fine-tuned BERT Base, reaches 99.83% test

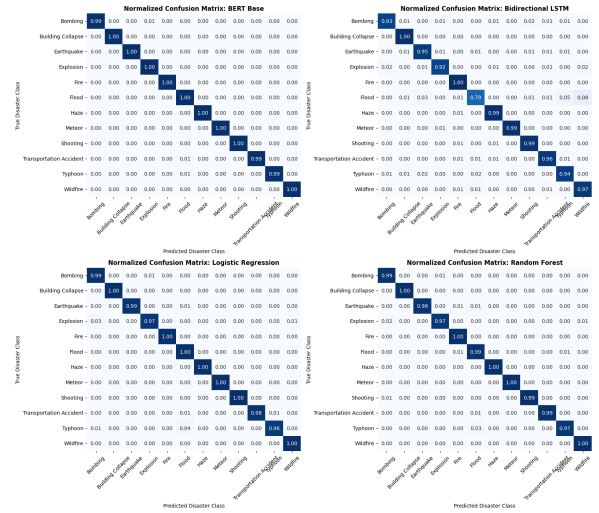


Figure 10: Confusion matrices: BERT Base, Bidirectional LSTM, Logistic Regression, SimpleRNN.

macro-F1.

Our main finding is that, although BERT and most other models scored really high, we should view these results carefully. Our EDA shows that many tweets contain the names of places, events, hashtags, etc. Thus, the model is essentially learning how to recognize these clues instead of general disaster-type language. We also found a separate problem. unidirectional GRU model collapsed to in all three tuning configurations and predicted only one class. Unidirectional LSTM performed worse as the model size got larger. But both architectures’ bidirectional versions had top tier scores – thus proving that reading the text both ways helped the sequential models.

Limitations. (1) We did not retrain the models after removing hashtags, names, and

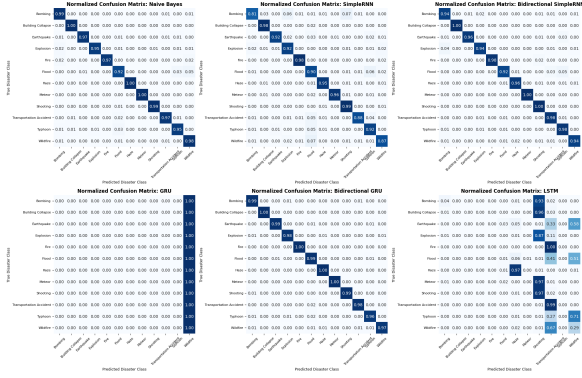


Figure 11: Confusion matrices: Naive Bayes, SimpleRNN, Bidirectional SimpleRNN, GRU, Bidirectional GRU, LSTM.

locations. Therefore, we cannot measure the difference between the models learning about disaster types and simply memorizing clues about specific events. (2) Some classes, such as Fire, Meteor, Bombing, Haze, and Shooting, contain tweets from only one event. This may have made those classes easier to identify. (3) The tweets were posted in 2012–2013. As social media is constantly changing and changing quickly, the social media language regarding maybe vastly different in present times. (4) A random post-level split may place near-duplicate messages, event-specific hashtags, or reports from the same crisis in both training and test data. The near-perfect scores therefore require confirmation with deduplication and event-disjoint evaluation. (5) A disaster can cascade into more disaster(s). A single-label classifier cannot represent a disaster that resulted in more disasters.

A Full Classification Reports

Full precision/recall/F1/support for the best model (BERT Base), the worst model (GRU), and representative mid-range models are given below. Random Forest’s full per-class report was not separately logged in the source notebook (only its aggregate accuracy/macro-F1, included in Table 3); this is noted here for transparency rather than omitted silently.

Class	P	R	F1	Supp.
Bombing	1.000	0.993	0.996	136
Building Collapse	1.000	1.000	1.000	125
Earthquake	1.000	1.000	1.000	154
Explosion	0.993	1.000	0.997	143
Fire	1.000	1.000	1.000	58
Flood	0.987	1.000	0.994	153
Haze	1.000	1.000	1.000	136
Meteor	1.000	1.000	1.000	132
Shooting	1.000	1.000	1.000	137
Transportation Accident	1.000	0.994	0.997	161
Typhoon	1.000	0.994	0.997	162
Wildfire	1.000	1.000	1.000	156
Macro avg	0.998	0.998	0.998	1653

Table 6: Full classification report: BERT Base (best model), test set.

Class	P	R	F1	Supp.
Bombing	0.000	0.000	0.000	136
Building Collapse	0.000	0.000	0.000	125
Earthquake	0.000	0.000	0.000	154
Explosion	0.000	0.000	0.000	143
Fire	0.000	0.000	0.000	58
Flood	0.000	0.000	0.000	153
Haze	0.000	0.000	0.000	136
Meteor	0.000	0.000	0.000	132
Shooting	0.000	0.000	0.000	137
Transportation Accident	0.000	0.000	0.000	161
Typhoon	0.000	0.000	0.000	162
Wildfire	0.094	1.000	0.173	156
Macro avg	0.008	0.083	0.014	1653

Table 7: Full classification report: GRU (worst model), test set. Predicted every test tweet as Wildfire.

Class	P	R	F1	Supp.
Bombing	0.625	0.037	0.069	136
Building Collapse	0.000	0.000	0.000	125
Earthquake	0.000	0.000	0.000	154
Explosion	0.000	0.000	0.000	143
Fire	0.000	0.000	0.000	58
Flood	0.200	0.013	0.025	153
Haze	0.830	0.971	0.895	136
Meteor	0.000	0.000	0.000	132
Shooting	0.209	0.971	0.343	137
Transportation Accident	0.316	0.994	0.479	161
Typhoon	0.000	0.000	0.000	162
Wildfire	0.139	0.295	0.189	156
Macro avg	0.193	0.273	0.167	1653

Table 8: Full classification report: LSTM, test set. Partial collapse across 4 classes at very low precision.

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